

Multi Tier Computing

Unit I

Introduction and Business Needs

1 Introduction and Business Needs

This unit introduces the business motivation and foundational concepts behind distributed computing and client/server architectures. It covers business opportunities, driving forces in information technology, major IT issues, right sizing, host and non-distributed computing, and the fundamentals of distributed and client/server computing.

2 The Business Opportunity

Businesses today use Information Technology (IT) to gain strategic, operational, and competitive advantages.

- Digital transformation enables new business models and revenue streams.
- Information systems provide competitive advantage through automation, speed, and intelligence.
- Data-driven decision making allows organizations to act quickly and accurately.
- Customer experience is enhanced through personalized services, mobile applications, and artificial intelligence.
- Operational efficiency is achieved through automation, optimization, and workflow improvements.

3 Driving Forces in Information Technology

Several forces push organizations to modernize and invest in IT.

- Globalization leading to increased competition and worldwide markets.

- Technological innovation such as Cloud Computing, Artificial Intelligence, Internet of Things, and Big Data.
- Demand for real-time information for instant insights and faster decisions.
- Automation and cost efficiency to reduce manual work and errors.
- User expectations for accessibility, speed, and personalization.

4 Major Issues in Information Technology

Despite the advantages, organizations face many challenges.

- Security threats and data privacy risks such as cyberattacks and data breaches.
- Integration challenges caused by coexistence of legacy and modern systems.
- Legacy systems that are expensive and difficult to modernize.
- Scalability and performance issues as data volume and number of users grow.
- High IT costs including infrastructure, training, and maintenance.
- Skill gaps in modern technologies such as cloud architecture and AI.

5 Cost–Performance Trade-offs

- High performance increases infrastructure cost.
- Low-cost systems may reduce response time.
- A balance between performance and budget is essential.
- Cost–performance trade-offs affect hardware, software, and networks.
- The goal is optimal performance at minimum cost.

Diagram: Cost vs Performance

Low Cost – Low Performance

High Cost – High Performance

6 Right Sizing

Right sizing refers to aligning IT resources exactly with business requirements.

- Avoiding over-sized systems that waste money.
- Avoiding under-sized systems that cause delays and failures.
- Using scalable models such as distributed computing and cloud computing.
- Ensuring cost-effective performance and future expansion capability.
- Selecting the correct mix of hardware, software, and human resources.

Diagram: Right Sizing

Under-Sized Right-Sized Over-Sized

7 Host and Non-Distributed Computing

7.1 Host (Centralized) Computing

- Based on mainframe systems where all processing occurs at one location.
- High reliability but low flexibility.
- Expensive and difficult to scale.
- Historically used in banks, government systems, and airline reservation systems.

7.2 Non-Distributed Computing

- All computing tasks occur on a single system.
- No distribution across multiple machines.
- Limited scalability.
- Easier to manage but difficult to adapt to modern requirements.
- Often replaced by distributed and client/server models.

8 Basics of Distributed Computing

- Computing is spread across multiple nodes.
- Improves scalability and performance.
- Enables resource sharing and parallel processing.
- Supports fault tolerance and high availability.

9 Decomposition Approaches

- Functional decomposition
- Data decomposition
- Domain-based decomposition
- Object-oriented decomposition

10 Layers vs Tiers

10.1 Layers

Layers represent logical separation of responsibilities.

- Presentation layer
- Business logic layer
- Data layer

10.2 Tiers

Tiers represent physical separation across machines.

- One-tier architecture
- Two-tier architecture
- Three-tier architecture
- N-tier architecture

11 Explanation of Layers

- Presentation Layer: User interface.
- Business Logic Layer: Processing and business rules.
- Data Layer: Data storage and retrieval.

12 Explanation of Tiers

- One-tier: All components on a single system.
- Two-tier: Client and Server.
- Three-tier: Client, Application Server, and Database.
- N-tier: Multiple distributed services.

13 Client/Server Computing

A client is any process that requests specific services from a server process. A server is a process that provides requested services for the client.

Client and server processes may reside on the same computer or on different computers connected through a network.

14 A Server for Every Client

14.1 File Server

- Provides clients access to records within files stored on the server.
- UNIX Network File Services (NFS) by Sun Microsystems.
- Microsoft Windows Map Drive.

14.2 Print Server

- Manages user access to shared output devices such as printers.

14.3 Database Server

- Provides data access in response to SQL requests.

14.4 Transaction Server

- Data and remote procedures reside on the server.
- Provides access to high-level functions.
- Implements efficient transaction processing.

14.5 Application Server

- Manages access to centralized application software.

14.6 Web Server

- Stores and retrieves Internet and intranet data.

14.7 Mail Server, Fax Server and Others

15 Fat and Thin Client

- Thin client performs minimal processing on the client side.
- Fat client performs a larger proportion of processing.

15.1 Fat Server

- Places more application functionality on the server.
- Opposite architecture is thin server.

16 Stateless and Stateful Servers

16.1 Stateless Server

- Treats each request as an independent transaction.
- Example: World Wide Web server.

16.2 Stateful Server

- Remembers previous client requests.
- Maintains information for incremental replies.
- Example: Remote file server.

17 Client/Server Functions

Client Functions

- Manages the user interface.
- Accepts and checks user input syntax.
- Processes application logic.
- Generates database requests and transmits them to the server.

Server Functions

- Accepts and processes database requests.
- Checks authorization.
- Ensures integrity constraints.
- Performs query and update processing.
- Provides concurrent database access.
- Provides recovery control.

18 Client/Server Topologies

- Multiple clients with a single server.
- Multiple clients with multiple servers.

19 Classification of Client/Server Systems

19.1 Two-Tier Client/Server Model

- Centralized two-tier representation.
- Business rules residing on the client.
- Business rules residing on the server.

19.2 Three-Tier Client/Server Model

- Presentation Logic
- Business Logic
- Database Management

19.3 N-Tier Client/Server Model

- GUI processing servers
- Business and transaction logic servers
- Data management servers

20 Client/Server Advantages

- Improved performance and reduced workload.
- Workstation independence.
- System interoperability.
- Scalability.
- Data integrity.
- Data accessibility.
- Simplified system administration.
- Resource sharing across diverse platforms.
- Location independence of data processing.

21 Client/Server Disadvantages

- Maintenance cost.
- Training cost.
- Hardware cost.
- Software cost.
- System complexity.